

Predictive State Plasticity

Status: independent high-risk proposal. The current submission-grade evidence is a 20-seed first-gate negative study: learned cue-GVF features do not yet become useful control state. Cue-decodability analysis has been added as a mechanism probe. Later candidate-feature replacement and feature-wise plasticity mechanisms remain future stages inside this research program, not completed positive evidence.

Abstract

Predictive knowledge is a central Alberta Plan idea, but a prediction is valuable to an agent only if it improves learning or control. The extended GVF predictive-state experiment is a negative result: trace memory and oracle memory solve a partially observable T-maze, while the learned GVF state remains near chance even with 20 seeds, 20000 online steps, and maze lengths up to 30. This proposal turns that failure into a sharper research program: how can a streaming agent with limited features select, adapt, and retain predictions that carry control-relevant hidden information?

Standalone Study Summary

This study asks whether learned predictions become useful state for control under partial observability. The RL problem is a T-maze with an early hidden cue and a delayed junction decision; raw observation is insufficient unless the agent retains cue information. The implemented comparison uses raw features, hand-coded trace memory, recurrent GVFs, redesigned cue GVFs, and oracle memory with online linear Sarsa control. The extended experiment varies maze length and representation mode over 20 seeds and 20000 online steps; the main metrics are trial accuracy, cue-alignment margin, GVF TD error, control TD error, and cue-decodability. The result is a useful negative gate: cue-GVF outputs contain weak cue information but do not improve control over chance, while trace and oracle memory solve the task. The next experiment must separate GVF output scaling, control utilization, and feature-selection/plasticity mechanisms.

Proposal Template Answers

Focused RL question: In a partially observable online control task, can learned predictions carry the hidden cue information needed for later action, or does the agent need a different state-construction mechanism? The present report answers this as a first-gate question about fixed learned GVF features, not as a completed feature-plasticity result.

Setting and testbed: The testbed is an aliased T-maze with an early binary cue and a delayed junction decision. It is intentionally sharper than a generic benchmark: raw observation is insufficient, trace/oracle memory can solve the task, and learned predictive state must prove that it carries useful hidden information.

Implemented comparison: The completed experiment compares raw observation, trace memory, recurrent GVF, cue GVF, and oracle memory with online linear Sarsa. Candidate-feature replacement, feature-wise step-size adaptation, feature ablation, and phase-change plasticity are staged next experiments, not current evidence.

Observation or figure that answers the question: The main evidence is trial accuracy by maze length, supported by cue-alignment margin and GVF TD error. A low GVF TD error or a small positive cue-alignment signal is not enough; the learned prediction must improve junction decisions over raw observation and move toward trace/oracle memory.

Compute need and fallback: The extended first-gate run is complete. If no further experiments are run, the honest fallback is to submit this as a negative/redesign proposal: current cue-GVF predictions do not yet become useful state, and the next work is diagnostic rather than performance polishing.

Independent Research Scope

This proposal is an independent representation and partial-observability study. The current scope is fixed predictive state under T-maze partial observability. Limited-budget feature replacement, generated traces, and feature-wise step-size adaptation are later stages inside the same research program, gated on whether a useful predictive feature can first be demonstrated. The present claim does not require those later mechanisms to succeed.

The proposal explicitly does not claim that GVFs fail in general. It claims that the current cue-GVF and recurrent-GVF constructions do not carry or expose the cue strongly enough for downstream linear control in this setting. This boundary is important because it turns a negative result into a design constraint rather than a broad dismissal of predictive knowledge.

Evidence Level

Evidence level: high-value negative gate with 20-seed extended evidence. The completed run covers maze lengths 8, 12, 20, and 30, five representation modes, and 20000 online steps per condition. The evidence is strong enough to say that the current learned GVF features do not become useful state in this T-maze.

The evidence is not a positive plasticity result. It now includes a cue-decodability probe, but it still does not include oracle-prediction control, GVF output normalization, feature ablations, phase switches, limited-budget feature replacement, or a fully specified feature-wise step-size adaptation baseline. Those missing experiments are not minor details; they are the necessary next gates for turning the negative result into a stronger research program.

Paper-Style Contribution And Claim Boundaries

The contribution is a negative but informative gate for predictive-state research. The report formulates a useful-prediction criterion under partial observability: a learned prediction must both carry hidden cue information and be usable by the control learner. By comparing raw observation, trace memory, GVFs, cue-GVFs, and oracle memory across maze lengths, the report separates task impossibility from representation failure.

The claim boundary is that the current GVF designs fail this gate; it is not a general theorem against GVFs or predictive knowledge. The academic value is the sharper next question: what information probe, output normalization, or feature-selection rule is required before learned predictions can compete with cheap trace memory?

Research Motivation

Partial observability makes current observation insufficient. A long-lived agent must build state from history, but the course constraints rule out deep recurrent networks and replay. GVFs offer a Core RL alternative: predictions about future signals can become features. The deeper issue is not whether predictive features can be computed, but whether they carry the hidden variable with enough strength and scaling for the control learner to use them. This report therefore treats useful predictive state as a measurable representation question before adding replacement or plasticity machinery.

Research Question

Current first-gate question:

> Before adding limited-budget replacement or feature-wise plasticity, can fixed learned GVF features carry the hidden cue well enough to improve online linear control over raw observation and approach cheap trace memory?

Longer staged-program question:

> When observations are aliased and the relevant memory timescale changes, what predictive question design and state-construction diagnostics help a streaming linear agent maintain useful state?

The present evidence answers only the first question. It is negative, and that negative result determines whether the larger staged program is worth pursuing.

Related Work

The Alberta Plan emphasizes GVFs, feature finding, and agent-state construction. Horde shows that many off-policy predictions can be learned in parallel. Useful-prediction work warns that prediction accuracy alone is not the right objective. Work on online agent-state construction motivates the staged plan: first test whether a prediction is informative and usable, then add capacity management and feature-wise plasticity only after the representation has passed that gate.

Useful local sources:

- - resources/alberta_plan_related/finding_useful_predictions_2111.11212.pdf
- - resources/alberta_plan_related/horde_lifelong_offpolicy_1206.6262.pdf
- - resources/alberta_plan_related/learning_agent_state_online_2112.15236.pdf
- - resources/alberta_plan_related/tidbd_1804.03334.pdf
- - resources/alberta_plan_related/streaming_deep_rl_finally_works_2410.14606.pdf
- - resources/alberta_plan_related/squeezing_more_from_stream_2602.09396.pdf

Research Method

The agent controls a partially observable task using linear action values over a fixed feature budget. Candidate features include:

- - current observation,
- - fixed decay traces,
- - GVF predictions with cue/junction/reward cumulants,
- - generated traces with learned decay parameters,
- - generated GVF questions selected by usefulness.

Learning variants:

- - Fixed feature set with ordinary TD/Sarsa.
- - Fixed feature set with normalized updates.
- - Planned limited-budget feature replacement based on downstream TD-error contribution.
- - Planned feature-wise step-size adaptation, treated as exploratory and only worth running after the fixed useful-GVF gate is understood.

Experimental Design

Primary environment:

- - Long aliased T-maze.
- - Cue appears at start and is hidden during the corridor.
- - Correct junction action depends on the cue.
- - Current implemented extended run keeps cue semantics fixed and varies corridor length. Phase-change experiments are planned but not yet implemented.

Secondary environment:

- - Planned, not yet implemented: sensor-chain prediction/control task where the relevant delay switches repeatedly.

Conditions:

- - Current implemented modes: raw observation, trace memory, recurrent GVF, cue GVF, and oracle memory.
- - Current implemented corridor lengths: 8, 12, 20, 30.
- - Planned, not yet implemented: fixed feature budgets 8, 16, 32.
- - Planned, not yet implemented: scheduled but unannounced phase switches.
- - Seeds 0-19 and 20000 online steps in the extended first-gate run.

Metrics:

- - Trial-end accuracy and reward.
- - GVF TD error.
- - Mutual-information proxy or correlation between features and hidden cue.
- - Recovery after phase switch.
- - Feature survival and replacement counts.
- - Per-feature alpha trajectories.
- - Downstream value loss with and without each feature group.

Experiment Design Rationale

The T-maze is used because it forces a clean separation between observation and state. If the agent does not retain the initial cue, the junction action is close to chance. Trace and oracle memory are included not as strawmen but as necessary controls: they show that the task is solvable under the course constraints and define how far the learned predictive feature is from a cheap memory baseline.

The current extended run is a gate rather than a full program. Maze length tests whether cue retention survives longer delays. Cue-alignment margin asks whether the learned feature contains any hidden-cue signal. Trial accuracy asks the stricter question of whether control can use that signal. GVF TD error is deliberately secondary because an accurate prediction can still be useless for control.

Expected Results And Failure Modes

Expected positive pattern: oracle memory and fixed trace memory should define upper bounds. Naive GVFs may fail unless their questions align with hidden cue information. Later replacement mechanisms should help only if utility is tied to downstream control; later feature-wise plasticity should reveal whether useful predictive features remain learnable after a phase switch.

Failure modes:

- - Prediction error may favor easy but useless cumulants.
- - Feature replacement may churn features and destabilize control.

- - Feature-wise plasticity may adapt step sizes without solving representation selection.
- - A hand-coded trace may remain a stronger baseline than learned GVFs; that is a valid negative result.

Interpretation Standard

The proposal should not claim "GVFs work" unless the learned predictions visibly carry the hidden variable and improve control over cheap memory baselines. A negative result is still valuable if it identifies a concrete design constraint for useful predictions.

Results

The completed extended first-gate run is:

experiments/alberta_core_rl/results/predictive_state_plasticity/20260709T024517Z_extended

This run compares raw observation, trace memory, old recurrent GVF, redesigned cue-GVF, and oracle memory across T-maze lengths 8, 12, 20, and 30, using seeds 0-19 and 20000 online steps per condition.

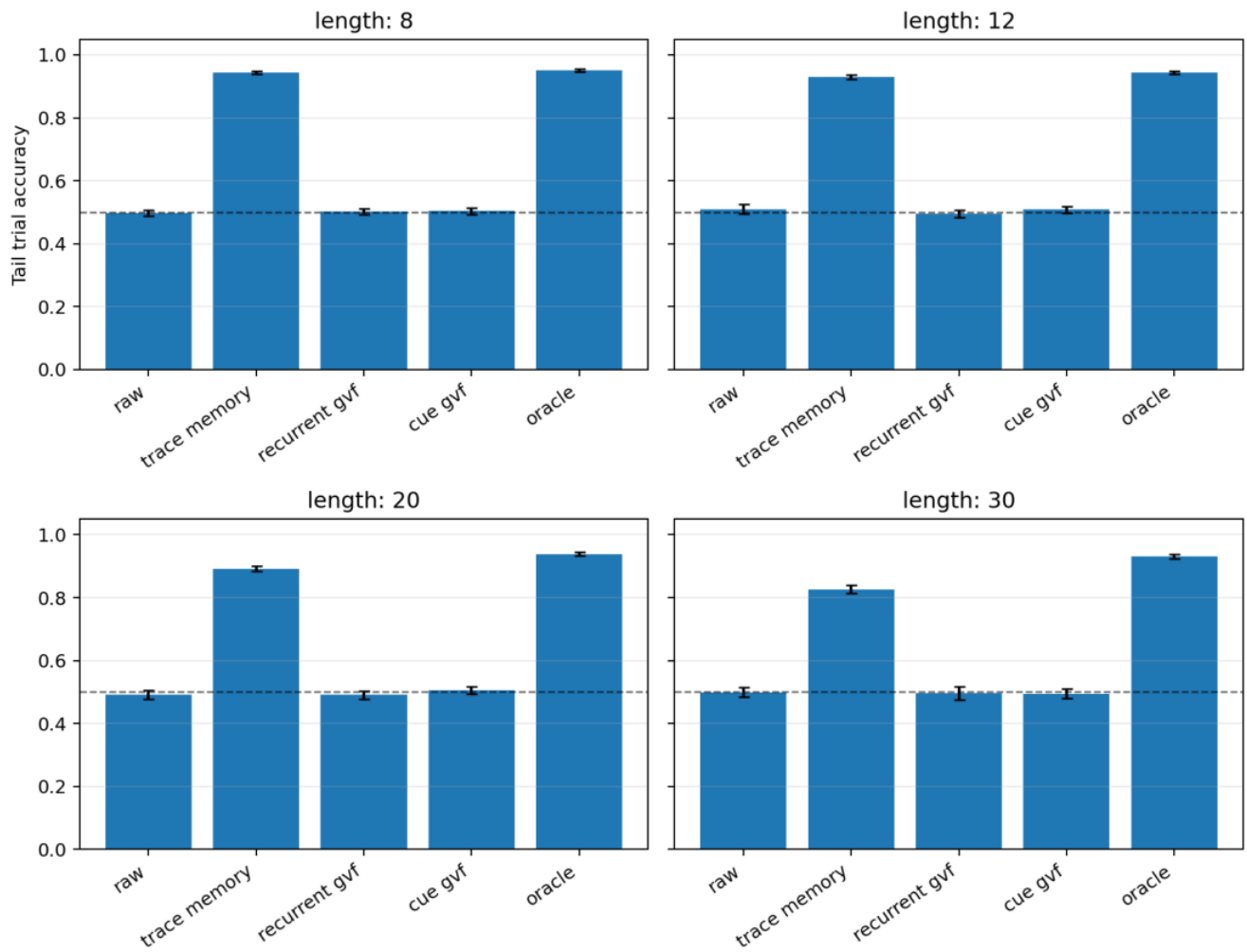
Main result:

- - Trace memory and oracle memory solve the shorter and medium lengths; trace memory remains far above chance even at length 30, though it drops to about 0.827 +/- 0.013.
- - Oracle memory remains near 0.931-0.950 across all tested lengths.
- - Raw observation and old recurrent GVF remain near chance.
- - Redesigned cue-GVF has some positive cue-alignment margin at shorter lengths, but the margin weakens as length grows.
- - Cue-GVF control accuracy remains near chance across all tested lengths: about 0.504, 0.508, 0.505, and 0.494 for lengths 8, 12, 20, and 30.

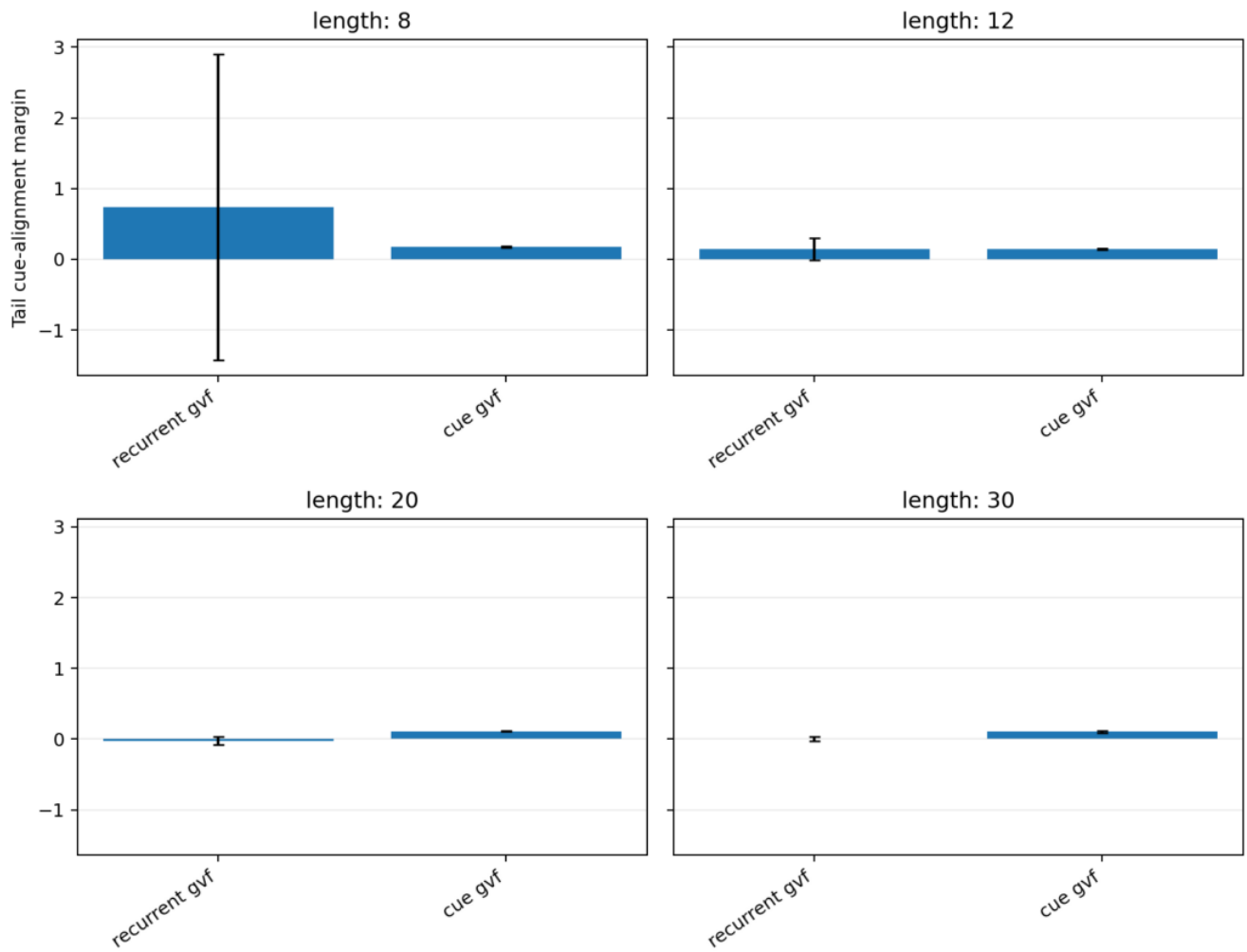
The added cue-decodability analysis separates information content from control usefulness. It computes whether each representation's late-corridor features predict the hidden cue before the junction. Cue-GVF features are above chance: tail cue-decoding accuracy is about 0.589, 0.589, 0.609, and 0.622 for maze lengths 8, 12, 20, and 30. Trace memory and oracle features are at 1.0, raw observation is at chance, and recurrent GVF remains around chance. This means the cue-GVF is not empty; it contains weak cue information that the downstream control learner does not turn into reliable decisions.

Interpretation: the redesigned GVF is a better diagnostic signal but still not sufficient state. The current result should be stated as "the learned GVF carries some cue information but does not expose it strongly enough for control," not as "GVFs cannot carry the cue." The next stage must separate information content from control utilization further by adding oracle-prediction controls, GVF output normalization, and then feature-plasticity/selection mechanisms.

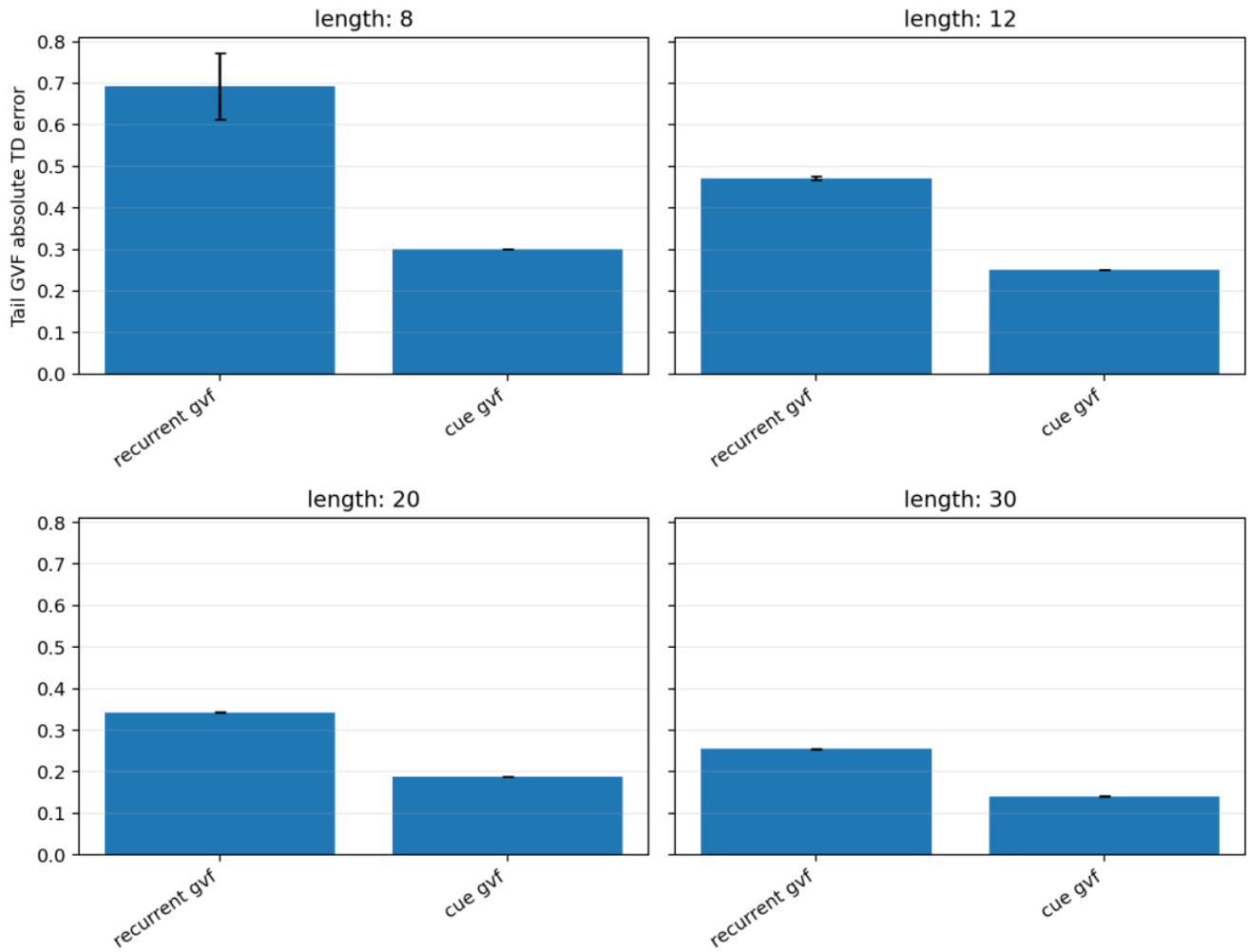
Main figures below use seed-tail condition summaries with 95% confidence intervals. They show the current negative gate more clearly than the earlier overloaded curve plots: cheap trace/oracle memory works, while learned GVF features remain near chance.



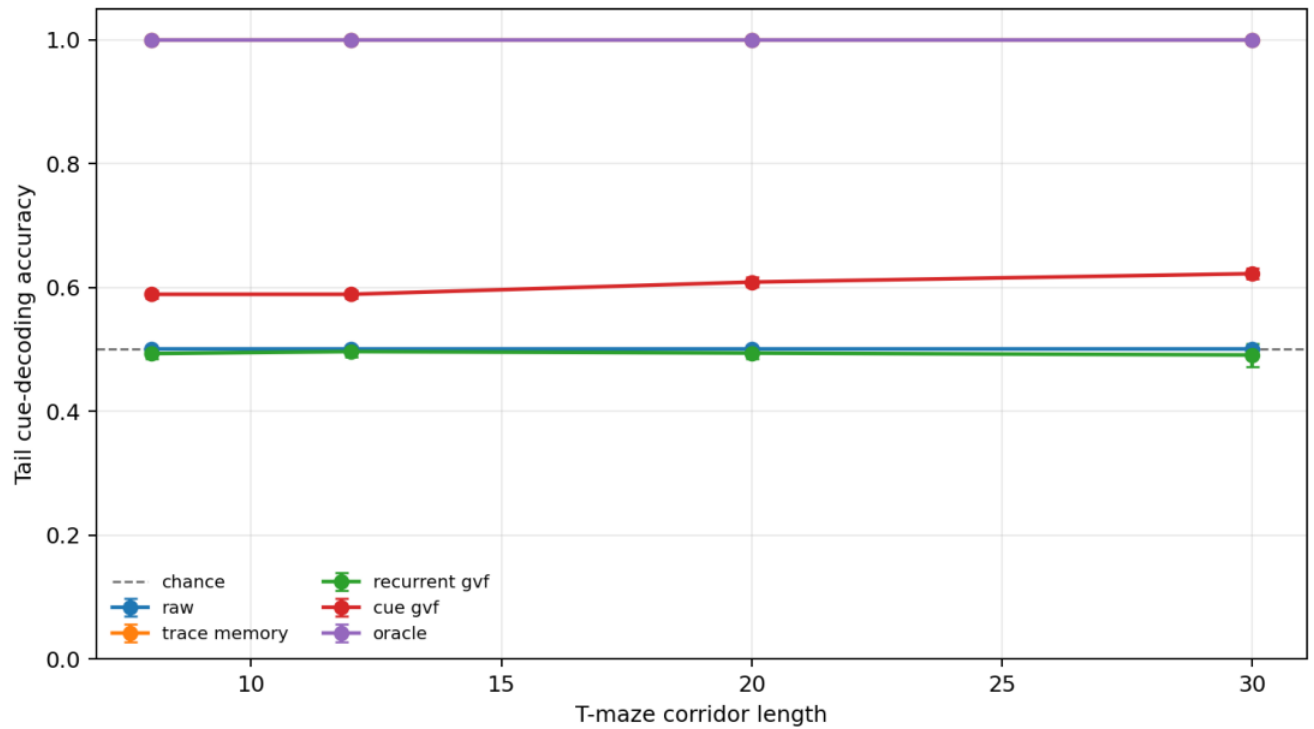
Tail trial accuracy by state construction and maze length.



Tail cue-alignment margin for GVF-based state constructions.



Tail GVF absolute TD error for GVF-based state constructions.



Tail cue-decoding accuracy by state construction and maze length.

Analysis

The central interpretation is that information content and control usefulness are separable. The cue-GVF is not empty: the cue-decodability probe shows above-chance hidden-cue information. Yet the control learner still behaves near chance, which means the learned prediction is either too weakly scaled, too noisy, or too poorly aligned with the downstream action-value update to serve as state. Trace and oracle memory prevent a trivial impossibility interpretation because they solve the same task under the same linear-control constraint. The negative result therefore points to a concrete redesign path: first make cue information decodable and usable, then test whether limited-budget feature selection or step-size plasticity can maintain that information over longer delays or phase changes.

Reviewer Critique And Revisions

Strict reviewer challenge: "The GVF result is negative, so why continue?" Response: the negative result is exactly the design signal. It says current GVF questions do not carry the cue; the next experiment must measure cue information and downstream usefulness directly.

Strict reviewer challenge: "This is too broad." Response: the proposal should be run in layers: first fixed useful GVF questions, then limited-budget selection, then step-size plasticity. The combined variant is not promoted until earlier layers pass.

Per-proposal audit matrix:

Reviewer angle	Critique	Action taken	Remaining risk
Alberta Plan	GVPs matter only if predictions become useful state. The report evaluates control accuracy, not only prediction error.	Current GVP questions still fail the useful-state gate.	
Partial observability	The task could be impossible without deep recurrence. Trace and oracle baselines solve the task. Learned prediction needs a stronger information probe.		
Representation	Cue alignment may exist but be unusable by control. Reports both cue-alignment margin and trial accuracy. Needs cue decodability and GVP output normalization.		
Plasticity	Generated features and feature-wise step-size adaptation are not yet tested in the full control loop. The report marks them as internal next-stage mechanisms rather than current evidence. A positive plasticity claim requires new experiments.		
Strict instructor	Do not sell a broad negative result as a broad GVP conclusion. Conclusion is limited to the current cue-GVP/recurrent-GVP designs. Better GVP questions may reverse the result.		

Threats To Validity

The current result is intentionally a gate, not a positive final claim. The T-maze is a controlled partial-observability test, but the learned GVP design is still weak and the control learner may not use small predictive margins effectively. The cue-decodability probe narrows the failure: the cue-GVP carries weak hidden-cue information, but that information does not become effective control state. The next version must separate output scaling, control utilization, oracle-prediction controls, and feature-selection/plasticity ablations.

Conclusion

This proposal remains ambitious and risky. Its value is that it refuses the shallow claim that "adding GVPs helps"; instead it asks whether predictions become useful state for control under partial observability. The first-gate pilot shows that the current cue-GVP has some cue-aligned information but is not yet a useful control state. That negative result is scientifically useful because it identifies the next design constraint: a predictive feature must be both informative about the hidden variable and usable by the downstream control update.

Reproduction

Current first-gate pilot:

```
cd /mnt/shared-storage-user/yupeng/Core-RL

PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
/data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/run_experiment.py \
--config experiments/alberta_core_rl/configs/predictive_state_plasticity/config_main.json
```

Extended sweep config:

```
cd /mnt/shared-storage-user/yupeng/Core-RL

PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
/data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/run_experiment.py \
--config experiments/alberta_core_rl/configs/predictive_state_plasticity/config_extended.json
```

Regenerate the report figures:

```
cd /mnt/shared-storage-user/yupeng/Core-RL

PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
/data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/plot_report_figures.py \
--kind predictive-state \
--result-dir experiments/alberta_core_rl/results/predictive_state_plasticity/20260709T024517Z_extended
```

Regenerate the cue-decodability summary and figure:

```
cd /mnt/shared-storage-user/yupeng/Core-RL

PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
/data/yupeng/conda_envs/core-rl/bin/python \
experiments/alberta_core_rl/scripts/analyze_predictive_state_decodability.py \
--result-dir experiments/alberta_core_rl/results/predictive_state_plasticity/20260709T024517Z_extended
```